

AIML Professional Portfolio Artifact

Training Generative AI Large Language Models Infrastructure, Resource Costs, and Training Pipeline Analysis

Name:	Benita Chinemerem
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1. Introduction

Large Language Models (LLMs) such as GPT-4 and Claude have transformed artificial intelligence by enabling systems to generate human-like text, summarize information, answer questions, and support reasoning and coding tasks. However, these systems require enormous infrastructure investments, large-scale datasets, powerful computational systems, and significant energy consumption during training and deployment. This portfolio artifact presents a visual slide-based infographic that explains the end-to-end training pipeline of modern generative AI systems and analyzes the major resources required to build and maintain them.

2. Description

This artifact is a professionally designed slide presentation that visually explains how large language models are trained and deployed. The presentation combines infrastructure analysis, visual workflow diagrams, and comparative resource analysis to help viewers understand the complexity and cost behind modern generative AI systems. The artifact focuses specifically on GPT-4 and Claude to demonstrate how infrastructure scale, compute power, alignment systems, and training costs influence model performance and capability.

3. Objective

- Clearly illustrate the key steps involved in training generative AI large language models.
 - Explain the major infrastructure components required for modern AI systems.
 - Analyze the primary resource categories associated with LLM development.
 - Compare GPT-4 and Claude training approaches and infrastructure investments.
 - Communicate complex AI concepts visually and professionally.
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4. Process

1. Reviewed assignment requirements and rubric expectations.
 2. Researched GPT-4 and Claude infrastructure using public technical references.
 3. Identified major resource categories including data, compute, energy, and alignment systems.
 4. Selected GPT-4 and Claude for deeper infrastructure analysis.
 5. Designed a slide-based infographic using portfolio-aligned visual formatting.
 6. Organized the presentation into workflow, infrastructure, cost, and impact sections.
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5. Key Concepts Covered

- Data Collection and preprocessing
 - Tokenization and transformer training
 - Distributed GPU infrastructure
 - Reinforcement Learning from Human Feedback (RLHF)
 - Energy consumption and operational cost
 - AI alignment and deployment systems
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6. Design Rationale

The presentation was intentionally designed as a slide-based infographic rather than a traditional text-heavy report. This approach improves visual communication and simplifies highly technical AI infrastructure concepts. GPT-4 and Claude were selected because they represent two important directions in generative AI development: scale-focused performance and safety-focused alignment. The design uses structured layouts, visual hierarchy, and portfolio-aligned branding to improve accessibility and professionalism.

7. Value Proposition

This artifact demonstrates my ability to understand modern AI infrastructure concepts, analyze generative AI training systems, organize technical information clearly, and communicate complex AI topics through professional visual storytelling. The project also demonstrates portfolio development and presentation design skills relevant to artificial intelligence and machine learning fields.

8. Relevance

This artifact is relevant because generative AI systems are becoming increasingly important across industries including healthcare, business, finance, education, and software engineering. Understanding the infrastructure, costs, and training requirements behind these systems is important for AI practitioners, researchers, and technology professionals.

9. Tools and Technologies Used

- Microsoft PowerPoint
 - GitHub Pages
 - OpenAI and Anthropic technical references
 - AI infrastructure research resources
 - Visual presentation design principles
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10. Conclusion

This artifact demonstrates that training advanced large language models requires far more than model architecture alone. Modern generative AI systems depend on distributed compute systems, massive datasets, energy-intensive training pipelines, alignment-focused optimization, and large-scale infrastructure investment. Through the comparison of GPT-4 and Claude, this artifact highlights how infrastructure decisions directly influence model capability, scalability, safety, and operational cost.
